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Administrar preferencias

Remote Services: Remote Desktop Protocol

Other sub-techniques of Remote Services (8)

Adversaries may use Valid Accounts to log into a computer using the Remote Desktop Protocol (RDP). The adversary may then perform actions as the logged-on user.

Remote desktop is a common feature in operating systems. It allows a user to log into an interactive session with a system desktop graphical user interface on a remote system. Microsoft refers to its implementation of the Remote Desktop Protocol (RDP) as Remote Desktop Services (RDS).[1]

Adversaries may connect to a remote system over RDP/RDS to expand access if the service is enabled and allows access to accounts with known credentials. Adversaries will likely use Credential Access techniques to acquire credentials to use with RDP. Adversaries may also use RDP in conjunction with the Accessibility Features or Terminal Services DLL for Persistence.[2]

ID: T1021.001

Sub-technique of: T1021

ⓘ

Tactic: Lateral Movement

ⓘ

Platforms: Windows

Contributors: Matthew Demaske, Adaptforward

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Procedure Examples

ID	Name	Description
C0063	2025 Poland Wiper Attacks	During the 2025 Poland Wiper Attacks, adversaries utilized RDP to log into jump hosts and then moved laterally to other victim devices to include a domain controller.[3]
G1030	Agrius	Agrius tunnels RDP traffic through deployed web shells to access victim environments via compromised accounts.[4] Agrius used the Plink tool to tunnel RDP connections for remote access and lateral movement in victim environments.[5]
G1024	Akira	Akira has used RDP for lateral movement.[6]
G0006	APT1	The APT1 group is known to have used RDP during operations.[7]
C0051	APT28 Nearest Neighbor Campaign	During APT28 Nearest Neighbor Campaign, APT28 used RDP for lateral movement.[8]
G0022	APT3	APT3 enables the Remote Desktop Protocol for persistence.[9] APT3 has also interacted with compromised systems to browse and copy files through RDP sessions.[10]
G0087	APT39	APT39 has been seen using RDP for lateral movement and persistence, in some cases employing the rdpwinst tool for mangement of multiple sessions.[11][12]
G0096	APT41	APT41 used RDP for lateral movement.[13][14] APT41 used NATBypass to expose local RDP ports on compromised systems to the Internet.[15]
G1023	APT5	APT5 has moved laterally throughout victim environments using RDP.[16]
G0143	Aquatic Panda	Aquatic Panda leveraged stolen credentials to move laterally via RDP in victim environments.[17]
G0001	Axiom	Axiom has used RDP during operations.[18]
G1043	BlackByte	BlackByte has used RDP to access other hosts within victim networks.[19][20]
G0108	Blue Mockingbird	Blue Mockingbird has used Remote Desktop to log on to servers interactively and manually copy files to remote hosts.[21]
C0015	C0015	During C0015, the threat actors used RDP to access specific network hosts of interest.[22]
C0018	C0018	During C0018, the threat actors opened a variety of ports to establish RDP connections, including ports 28035, 32467, 41578, and 46892.[23]
C0032	C0032	During the C0032 campaign, TEMP.Veles utilized RDP throughout an operation.[24]
S0030	Carbanak	Carbanak enables concurrent Remote Desktop Protocol (RDP) sessions.[25]
G0114	Chimera	Chimera has used RDP to access targeted systems.[26]
G0080	Cobalt Group	Cobalt Group has used Remote Desktop Protocol to conduct lateral movement.[27]
S0154	Cobalt Strike	Cobalt Strike can start a VNC-based remote desktop server and tunnel the connection through the already established C2 channel.[28][29]
C0029	Cutting Edge	During Cutting Edge, threat actors used RDP with compromised credentials for lateral movement.[30]

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ID	Name	Description
G0051	FIN10	FIN10 has used RDP to move laterally to systems in the victim environment.[33]
G1016	FIN13	FIN13 has remotely accessed compromised environments via Remote Desktop Services (RDS) for lateral movement.[34]
G0037	FIN6	FIN6 used RDP to move laterally in victim networks.[35][36]
G0046	FIN7	FIN7 has used RDP to move laterally in victim environments.[37]
G0061	FIN8	FIN8 has used RDP for lateral movement.[38]
G0117	Fox Kitten	Fox Kitten has used RDP to log in and move laterally in the target environment.[39][40]
G1001	HEXANE	HEXANE has used remote desktop sessions for lateral movement.[41]
C0038	HomeLand Justice	During HomeLand Justice, threat actors primarily used RDP for lateral movement in the victim environment.[42][43]
S0434	Imminent Monitor	Imminent Monitor has a module for performing remote desktop access.[44]
G1032	INC Ransom	INC Ransom has used RDP to move laterally.[45][46][47][48]
G0119	Indrik Spider	Indrik Spider has used RDP for lateral movement.[49]
S0283	jRAT	jRAT can support RDP control.[50]
G0094	Kimsuky	Kimsuky has used RDP for direct remote point-and-click access.[51]
S0250	Koadic	Koadic can enable remote desktop on the victim's machine.[52]
G0032	Lazarus Group	Lazarus Group malware SierraCharlie uses RDP for propagation.[53][54]
G0065	Leviathan	Leviathan has targeted RDP credentials and used it to move through the victim environment.[55]
G0059	Magic Hound	Magic Hound has used Remote Desktop Services to copy tools on targeted systems.[56][57]
G1051	Medusa Group	Medusa Group has used RDP to conduct lateral movement and exfiltrate data.[58] Medusa Group has also utilized the Windows executable `mstsc.exe` for RDP activities through the command `mstsc.exe /v:{hostname/ip}`.[58]
G0045	menuPass	menuPass has used RDP connections to move across the victim network.[59][60]
G1054	MirrorFace	MirrorFace has used RDP to exfiltrate files of interest.[61]
G1059	NIMBUS MANTICORE	NIMBUS MANTICORE has utilized RDP to log into victim devices using compromised user credentials.[62]
S0385	njRAT	njRAT has a module for performing remote desktop access.[63][64]
G0049	OilRig	OilRig has used Remote Desktop Protocol for lateral movement. The group has also used tunneling tools to tunnel RDP into the environment.[65][66][14][67][67]
C0061	Operation Digital Eye	During Operation Digital Eye, threat actors moved laterally using RDP.[68]

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ID	Name	Description
S0583	Pysa	Pysa has laterally moved using RDP connections.[71]
S0262	QuasarRAT	QuasarRAT has a module for performing remote desktop access.[72][73]
S1187	reGeorg	reGeorg can be used to tunnel RDP connections.[74]
S0379	Revenge RAT	Revenge RAT has a plugin to perform RDP access.[75]
G1045	Salt Typhoon	Salt Typhoon has opened and conducted operations utilizing RDP within victim environments.[76]
G1015	Scattered Spider	Scattered Spider has used RDP to enable lateral movement.[77]
S0461	SDBbot	SDBbot has the ability to use RDP to connect to victim's machines.[78]
S0382	ServHelper	ServHelper has commands for adding a remote desktop user and sending RDP traffic to the attacker through a reverse SSH tunnel.[79]
G0091	Silence	Silence has used RDP for lateral movement.[80]
C0024	SolarWinds Compromise	During the SolarWinds Compromise, APT29 used RDP sessions from public-facing systems to internal servers.[81]
S9051	The Gentlemen Ransomware	The Gentlemen Ransomware has enabled Remote Desktop Protocol within victim environments.[82]
G1055	VOID MANTICORE	VOID MANTICORE has used RDP to move laterally within the victim environment.[83]
G1017	Volt Typhoon	Volt Typhoon has moved laterally to the Domain Controller via RDP using a compromised account with domain administrator privileges.[84]
S0670	WarzoneRAT	WarzoneRAT has the ability to control an infected PC using RDP.[85]
G0102	Wizard Spider	Wizard Spider has used RDP for lateral movement and to deploy ransomware interactively.[86][87][88][89]
S0350	zwShell	zwShell has used RDP for lateral movement.[90]
S0412	ZxShell	ZxShell has remote desktop functionality.[91]

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Mitigations

ID	Mitigation	Description
M1047	Audit	Audit the Remote Desktop Users group membership regularly. Remove unnecessary accounts and groups from Remote Desktop Users groups.
M1042	Disable or Remove Feature or Program	Disable the RDP service if it is unnecessary.
M1035	Limit Access to Resource Over Network	Use remote desktop gateways.
M1032	Multi-factor Authentication	Use multi-factor authentication for remote logins.[92]
M1030	Network Segmentation	Do not leave RDP accessible from the internet. Enable firewall rules to block RDP traffic between network security zones within a network.
M1028	Operating System Configuration	Change GPOs to define shorter timeouts sessions and maximum amount of time any single session can be active. Change GPOs to specify the maximum amount of time that a disconnected session stays active on the RD session host server.[93]
M1026	Privileged Account Management	Consider removing the local Administrators group from the list of groups allowed to log in through RDP.
M1018	User Account Management	Limit remote user permissions if remote access is necessary.

Detection Strategy

ID	Name	Analytic ID	Analytic Description
DET0327	Multi-event Detection Strategy for RDP-Based Remote Logins and Post-Access Activity	AN0931	Remote Desktop (RDP) logon by a user followed by unusual process execution, file access, or lateral movement activity within a short timeframe.

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References

1. Microsoft. (n.d.). Remote Desktop Services. Retrieved June 1, 2016.
2. Alperovitch, D. (2014, October 31). Malware-Free Intrusions. Retrieved November 17, 2024.
3. CERT Polska. (2026, January 30). Energy Sector Incident Report – 29 December. Retrieved April 22, 2026.
4. Amitai Ben & Shushan Ehrlich. (2021, May). From Wiper to Ransomware: The Evolution of Agrius. Retrieved May 21, 2024.
5. Or Chechik, Tom Fakterman, Daniel Frank & Assaf Dahan. (2023, November 6). Agonizing Serpens (Aka Agrius) Targeting the Israeli Higher Education and Tech Sectors. Retrieved May 22, 2024.
6. Nutland, J. and Szeliga, M. (2024, October 21). Akira ransomware continues to evolve. Retrieved December 10, 2024.
7. FireEye Labs. (2014, May 20). The PLA and the 8:00am-5:00pm Work Day: FireEye Confirms DOJ's Findings on APT1 Intrusion Activity. Retrieved November 17, 2024.
8. Koessel, Sean. Adair, Steven. Lancaster, Tom. (2024, November 22). The Nearest Neighbor Attack: How A Russian APT Weaponized Nearby Wi-Fi Networks for Covert Access. Retrieved February 25, 2025.
9. valsmith. (2012, September 21). More on APTSim. Retrieved September 28, 2017.
10. Glyer, C. (2018, April 14). @cglyer Status Update. Retrieved September 12, 2024.
11. Hawley et al. (2019, January 29). APT39: An Iranian Cyber Espionage Group Focused on Personal Information. Retrieved February 19, 2019.
12. Rusu, B. (2020, May 21). Iranian Chafer APT Targeted Air Transportation and Government in Kuwait and Saudi Arabia. Retrieved May 22, 2020.
13. Fraser, N., et al. (2019, August 7). Double DragonAPT41, a dual espionage and cyber crime operation APT41. Retrieved September 23, 2019.
14. Crowdstrike. (2020, March 2). 2020 Global Threat Report. Retrieved December 11, 2020.
15. DCSO CyTec Blog. (2022, December 24). APT41 — The spy who failed to encrypt me. Retrieved June 13, 2024.
16. Perez, D. et al. (2021, May 27). Re-Checking Your Pulse: Updates on Chinese APT Actors Compromising Pulse Secure VPN Devices. Retrieved February 5, 2024.
17. CrowdStrike. (2023). 2022 Falcon OverWatch Threat Hunting Report. Retrieved May 20, 2024.
18. Novetta. (n.d.). Operation SMN: Axiom Threat Actor Group Report. Retrieved November 12, 2014.
19. Microsoft Incident Response. (2023, July 6). The five-day job: A BlackByte ransomware intrusion case study. Retrieved December 16, 2024.
20. James Nutland, Craig Jackson, Terryn Valikodath, & Brennan Evans. (2024, August 28). BlackByte blends tried-and-true tradecraft with newly disclosed vulnerabilities to support ongoing attacks. Retrieved December 16, 2024.
21. Lambert, T. (2020, May 7). Introducing Blue Mockingbird.
48. Carvey, H. (2024, May 1). LOLBin to INC Ransomware. Retrieved June 5, 2024.
49. Mandiant Intelligence. (2022, June 2). To HADES and Back: UNC2165 Shifts to LOCKBIT to Evade Sanctions. Retrieved July 29, 2024.
50. Kamluk, V. & Gostev, A. (2016, February). Adwind - A Cross-Platform RAT. Retrieved April 23, 2019.
51. ASERT team. (2018, December 5). STOLEN PENCIL Campaign Targets Academia. Retrieved February 5, 2019.
52. Magius, J., et al. (2017, July 19). Koadic. Retrieved September 27, 2024.
53. Novetta Threat Research Group. (2016, February 24). Operation Blockbuster: Unraveling the Long Thread of the Sony Attack. Retrieved February 25, 2016.
54. Novetta Threat Research Group. (2016, February 24). Operation Blockbuster: Remote Administration Tools & Content Staging Malware Report. Retrieved March 16, 2016.
55. Plan, F., et al. (2019, March 4). APT40: Examining a China-Nexus Espionage Actor. Retrieved March 18, 2019.
56. DFIR Report. (2022, March 21). APT35 Automates Initial Access Using ProxyShell. Retrieved May 25, 2022.
57. DFIR Report. (2021, November 15). Exchange Exploit Leads to Domain Wide Ransomware. Retrieved January 5, 2023.
58. Cybersecurity and Infrastructure Security Agency. (2025, March 12). AA25-071A #StopRansomware: Medusa Ransomware. Retrieved October 15, 2025.
59. PwC and BAE Systems. (2017, April). Operation Cloud Hopper. Retrieved April 5, 2017.
60. US District Court Southern District of New York. (2018, December 17). United States v. Zhu Hua Indictment. Retrieved December 17, 2020.
61. Trend Micro. (2024, November 19). Spot the Difference: Earth Kasha's New LODEINFO Campaign And The Correlation Analysis With The APT10 Umbrella. Retrieved April 17, 2026.
62. El-Banna, M., et al. (2025, November 17). Frontline Intelligence: Analysis of UNC1549 TTPs, Custom Tools, and Malware Targeting the Aerospace and Defense Ecosystem. Retrieved September 28, 2026.
63. Fidelis Cybersecurity. (2013, June 28). Fidelis Threat Advisory #1009: "njRAT" Uncovered. Retrieved June 4, 2019.
64. Global Research & Analysis Team, Kaspersky. (2024, August 19). BlindEagle flying high in Latin America. Retrieved April 16, 2026.
65. Unit42. (2016, May 1). Evasive Serpens Unit 42 Playbook Viewer. Retrieved February 6, 2023.
66. Davis, S. and Caban, D. (2017, December 19). APT34 - New Targeted Attack in the Middle East. Retrieved December 20, 2017.
67. Symantec Threat Hunter Team. (2023, October 19). Crambus: New Campaign Targets Middle Eastern Government. Retrieved November 27, 2024.
68. Aleksandar Milenkoski, Luigi Martire. (2024, December 10). Operation Digital Eye | Chinese APT Compromises Critical Digital Infrastructure via Visual Studio Code Tunnels. Retrieved February 27, 2025.

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24. Miller, S, et al. (2019, April 10). TRITON Actor TTP Profile, Custom Attack Tools, Detections, and ATT&CK Mapping. Retrieved April 16, 2019.
25. Bennett, J., Vengerik, B. (2017, June 12). Behind the CARBANAK Backdoor. Retrieved June 11, 2018.
26. Cycraft. (2020, April 15). APT Group Chimera - APT Operation Skeleton key Targets Taiwan Semiconductor Vendors. Retrieved August 24, 2020..
27. Matveeva, V. (2017, August 15). Secrets of Cobalt. Retrieved October 10, 2018.
28. Strategic Cyber LLC. (2017, March 14). Cobalt Strike Manual. Retrieved May 24, 2017.
29. Cybereason. (2022, August 17). Bumblebee Loader – The High Road to Enterprise Domain Control. Retrieved August 29, 2022.
30. Meltzer, M. et al. (2024, January 10). Active Exploitation of Two Zero-Day Vulnerabilities in Ivanti Connect Secure VPN. Retrieved February 27, 2024.
31. Kujawa, A. (2018, March 27). You dirty RAT! Part 1: DarkComet. Retrieved November 6, 2018.
32. US-CERT. (2018, March 16). Alert (TA18-074A): Russian Government Cyber Activity Targeting Energy and Other Critical Infrastructure Sectors. Retrieved June 6, 2018.
33. FireEye iSIGHT Intelligence. (2017, June 16). FIN10: Anatomy of a Cyber Extortion Operation. Retrieved November 17, 2024.
34. Ta, V., et al. (2022, August 8). FIN13: A Cybercriminal Threat Actor Focused on Mexico. Retrieved February 9, 2023.
35. FireEye Threat Intelligence. (2016, April). Follow the Money: Dissecting the Operations of the Cyber Crime Group FIN6. Retrieved November 17, 2024.
36. McKeague, B. et al. (2019, April 5). Pick-Six: Intercepting a FIN6 Intrusion, an Actor Recently Tied to Ryuk and LockerGoga Ransomware. Retrieved April 17, 2019.
37. Loui, E. and Reynolds, J. (2021, August 30). CARBON SPIDER Embraces Big Game Hunting, Part 1. Retrieved September 20, 2021.
38. Elovitz, S. & Ahl, I. (2016, August 18). Know Your Enemy: New Financially-Motivated & Spear-Phishing Group. Retrieved February 26, 2018.
39. CISA. (2020, September 15). Iran-Based Threat Actor Exploits VPN Vulnerabilities. Retrieved December 21, 2020.
40. ClearSky. (2020, December 17). Pay2Key Ransomware – A New Campaign by Fox Kitten. Retrieved December 21, 2020.
41. SecureWorks 2019, August 27 LYCEUM Takes Center Stage in Middle East Campaign Retrieved. 2019/11/19
42. CISA. (2022, September 23). AA22-264A Iranian State Actors Conduct Cyber Operations Against the Government of Albania. Retrieved August 6, 2024.
43. MSTIC. (2022, September 8). Microsoft investigates Iranian attacks against the Albanian government. Retrieved August 6, 2024.
44. QiAnXin Threat Intelligence Center. (2019, February 18). APT-C-36: Continuous Attacks Targeting Colombian Government Institutions and Corporations. Retrieved May 5, 2020.
45. Cybereason Security Research Team. (2023, November 20). Threat Alert: INC Ransomware. Retrieved June 5, 2024.
46. Team Huntress. (2023, August 11). Investigating New INC Ransom Group Activity. Retrieved June 5, 2024.
72. MaxXor. (n.d.). QuasarRAT. Retrieved July 10, 2018.
73. Meltzer, M, et al. (2018, June 07). Patchwork APT Group Targets US Think Tanks. Retrieved July 16, 2018.
74. FortiGard Labs. (2019, March 12). ReGeorg.HTTP.Tunnel. Retrieved December 3, 2024.
75. Livelli, K, et al. (2018, November 12). Operation Shaheen. Retrieved May 1, 2019.
76. CISA. (2025, September 3). Countering Chinese State-Sponsored Actors Compromise of Networks Worldwide to Feed Global Espionage System. Retrieved September 22, 2026.
77. Mandiant Incident Response. (2025, May 6). Defending Against UNC3944: Cybercrime Hardening Guidance from the Frontlines. Retrieved October 13, 2025.
78. Schwarz, D. et al. (2019, October 16). TA505 Distributes New SDBbot Remote Access Trojan with Get2 Downloader. Retrieved May 29, 2020.
79. Schwarz, D. and Proofpoint Staff. (2019, January 9). ServHelper and FlawedGrace - New malware introduced by TA505. Retrieved May 28, 2019.
80. Group-IB. (2018, September). Silence: Moving Into the Darkside. Retrieved May 5, 2020.
81. CrowdStrike. (2022, January 27). Early Bird Catches the Wormhole: Observations from the StellarParticle Campaign. Retrieved February 7, 2022.
82. Santos, J., et al. (2025, September 9). Unmasking The Gentlemen Ransomware: Tactics, Techniques, and Procedures Revealed . Retrieved September 18, 2026.
83. Check Point Research. (2026, March 12). “Handala Hack” – Unveiling Group’s Modus Operandi. Retrieved April 20, 2026.
84. CISA et al.. (2024, February 7). PRC State-Sponsored Actors Compromise and Maintain Persistent Access to U.S. Critical Infrastructure. Retrieved May 15, 2024.
85. Harakhavik, Y. (2020, February 3). Warzone: Behind the enemy lines. Retrieved December 17, 2021.
86. John, E. and Carvey, H. (2019, May 30). Unraveling the Spiderweb: Timelining ATT&CK Artifacts Used by GRIM SPIDER. Retrieved May 12, 2020.
87. DHS/CISA. (2020, October 28). Ransomware Activity Targeting the Healthcare and Public Health Sector. Retrieved October 28, 2020.
88. The DFIR Report. (2020, November 5). Ryuk Speed Run, 2 Hours to Ransom. Retrieved November 6, 2020.
89. Shilko, J., et al. (2021, October 7). FIN12: The Prolific Ransomware Intrusion Threat Actor That Has Aggressively Pursued Healthcare Targets. Retrieved June 15, 2023.
90. McAfee® Foundstone® Professional Services and McAfee Labs™. (2011, February 10). Global Energy Cyberattacks: “Night Dragon”. Retrieved February 19, 2018.
91. Allievi, A., et al. (2014, October 28). Threat Spotlight: Group 72, Opening the ZxShell. Retrieved September 24, 2019.
92. Berkeley Security, University of California. (n.d.). Securing Remote Desktop for System Administrators. Retrieved November 4, 2014.
93. Microsoft. (n.d.). Configure Timeout and Reconnection Settings for Remote Desktop Services Sessions. Retrieved December 11, 2017.

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